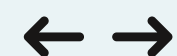
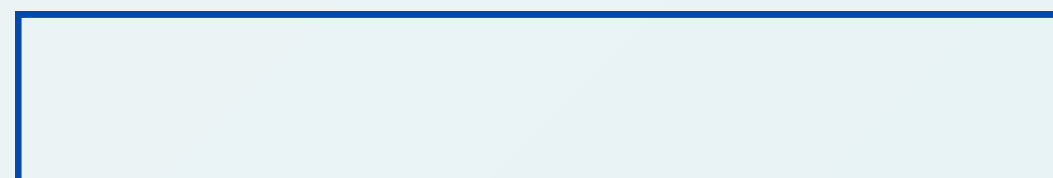




PROJECT OVERVIEW & CLINICAL MOTIVATION

Style-Invariant Multi-Task Triage of
Post-C-Section Portal Messages



PROJECT OVERVIEW & CLINICAL MOTIVATION

The Clinical Problem:

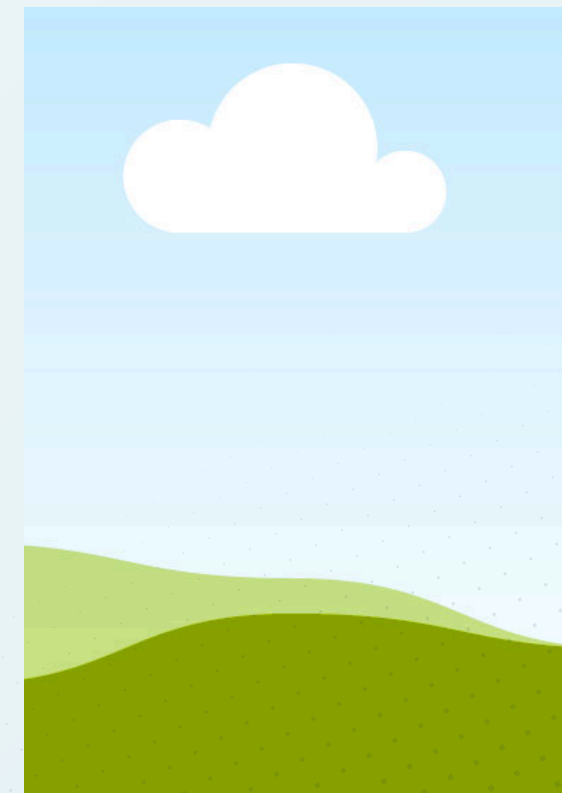
Asynchronous patient portal messages are the primary communication line post-C-section, but they suffer from high linguistic noise (Anxiety, Low Medical Literacy, Understatement) that distorts risk signals.

The HIPAA Wall:

Authentic data is inaccessible due to PHI/privacy restrictions.

Our Objective:

Developing a controlled, attribute-conditioned synthetic data pipeline (N=3,000) to build a robust triage benchmark and evaluate style-invariance across clinical transformers and foundational LLMs.





MULTI-TASK FRAMEWORK & MEDICAL GROUNDING

01.

Task A (Core ML Benchmark):

8-dimensional binary multi-label classification tracking complications: [infection, wound, bleeding, severe_pain, urinary, respiratory, leg_clot, routine_recovery].

02.

Task B (Severity Routing):

Rule-based Clinical Severity Mapping to triage priority levels (Low, Medium, High).

03.

Task C (Explainability):

Extractive text Evidence Spans for clinical auditability.

Medical Grounding:

Logic based on official NHS Post-Incision Discharge Guidelines (2024) and Mayo Clinic Postpartum Protocols (e.g., Fever defined strictly as $38 \leq ^\circ\text{C}$).

CONTROLLED PRE-SAMPLING STRATEGY & DATASET SPLIT

=



The Pipeline:

Moving from Label-to-Data (Controlled generation using a fixed meta-attribute dictionary).

Persona Confounders Embedded:

Speaker (Patient/Caregiver), Anxiety (Low/High), Medical Literacy (Low/High), Understatement (Low/High), Message Length (Short/Medium/Long).

Dataset Splits (N=3,000):

- Train / Validation / Test A (Controlled): 50% Routine vs. 50% Symptoms to optimize Recall and balance class distribution.
- Test B (Realistic Clinical Stress Test): 80% Routine vs. 20% Symptom heavy, accurately simulating an actual live hospital deployment.

PILOT PHASE EXECUTION & ERROR ANALYSIS (THE KEY FINDING)

Pilot Setup:

Execution of a manual validation cycle using foundational models (gpt-4o/Claude-3.5-Sonnet) based on our system prompt blueprint.

Qualitative Success:

Models successfully adopted complex personas, capturing implicit warnings under high understatement conditions (e.g., caregiver minimizing a DVT sign).

Structural Error Caught (The JSON Bug):

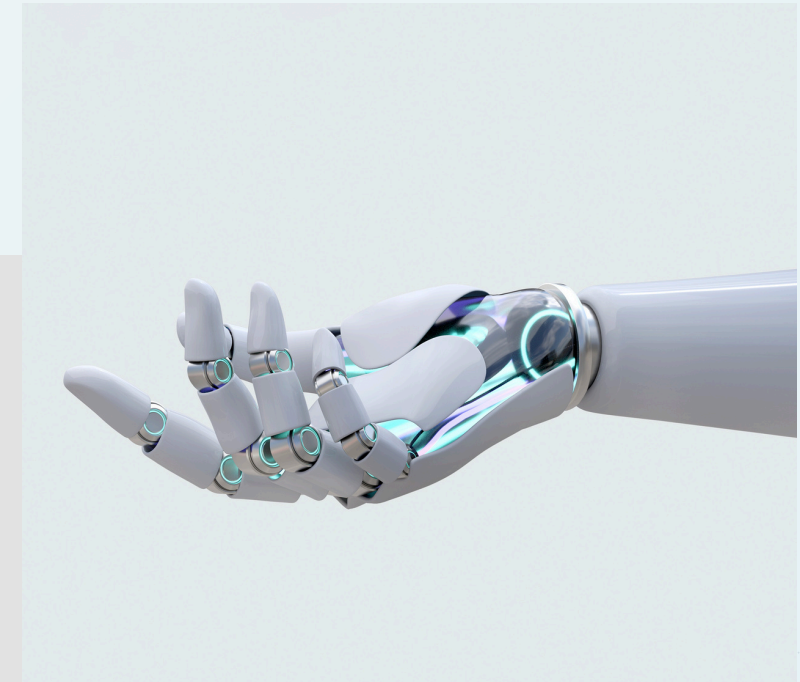
- The Issue:
The LLM hallucinated a static key "active_label_name" inside the evidence_spans block instead of rendering the dynamic label name requested.
- The Engineering Fix:
Implementation of an automated strict programmatic validation layer to reject and regenerate malformed JSON objects automatically during production.

BENCHMARKING MATRIX & NEXT STEPS



Benchmarking Architectures (Side-by-Side):

1. Heuristic Baseline: TF-IDF + Logistic Regression (Binary Relevance).
2. Contextual General Encoder: RoBERTa-Base (Full fine-tuning).
3. Domain-Specific Clinical Encoder: Bio_ClinicalBERT (MIMIC-III pre-trained).
4. Foundational Frontier LLMs: Zero-Shot/Few-Shot gpt-4o-mini and Llama-3-8B.



Core Experiment:

Style Ablation Study (Training with and without style diversity, followed by targeted sub-group stress testing).

Next Immediate Milestones:

Automating the full data generation loop using bulk parsing scripts and executing Baseline training.



